BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI
Hyderabad Campus
SECOND SEMESTER 2025-2026

Mathematical Foundations for Data Science

Tutorial Sheet - 2

1. Find the trace and determinant of the matrix.

   (i) $\begin{bmatrix} 5 & 2 & 2 \\ 1 & 0 & 3 \\ 2 & 4 & 7 \end{bmatrix}$

   (ii) $\begin{bmatrix} 10 & 3 & 3 \\ 0 & -1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

2. Find the characteristic polynomial, eigenvalues and eigenvectors of the following matrices

   (i) $\begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

   (ii) $\begin{bmatrix} 1 & 1 & 1 \\ -1 & -3 & -3 \\ 2 & 4 & 4 \end{bmatrix}$

3. Find the Cholesky factorization of the matrix $\begin{bmatrix} 4 & 12 & -16 \\ 12 & 37 & -43 \\ -16 & -43 & 98 \end{bmatrix}$

4. Compute the eigendecomposition of the following matrices.

   (i) $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

   (ii) $\begin{bmatrix} 3 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix}$

5. Compute the singular value decomposition (SVD) of the following matrices.

   (i) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$

   (ii) $\begin{bmatrix} 3 & 1 & 1 & 0 \\ 1 & 2 & 2 & -1 \end{bmatrix}$